

CompSci 234/NetSys 210
Advanced Topics in Networking
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Unit 12: Wireless Networks

1

Cheng-Hsin Hsu (chsu@cs.nthu.edu.tw)

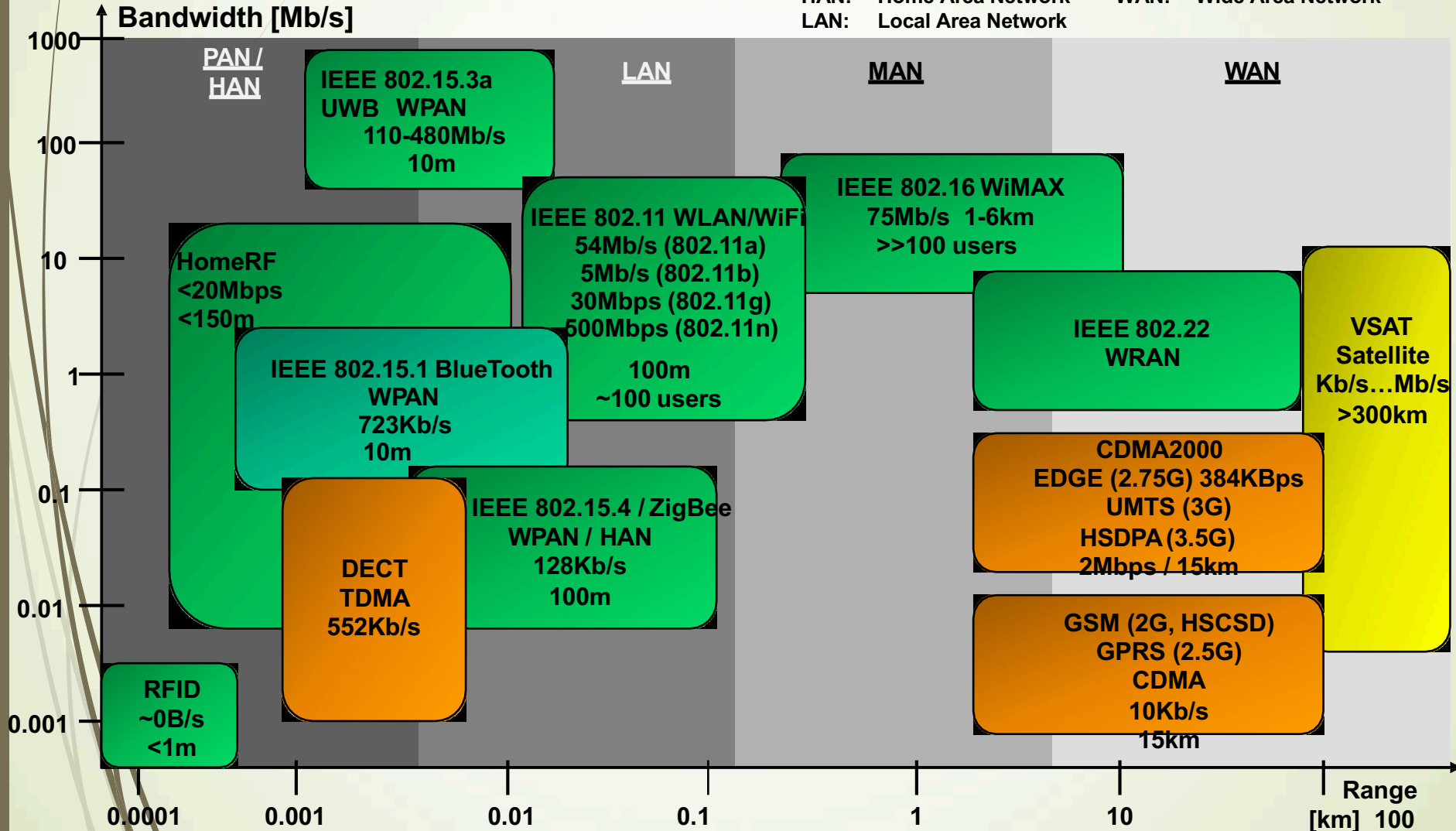
Some Slide adopted from Peter Egli 's and Prof. Venkatasubramanian's materials

Different Wireless Networks

2

Q: What is the difference between Green and Orange boxes?

PAN: Personal Area Network MAN: Metropolitan Area Network
HAN: Home Area Network WAN: Wide Area Network
LAN: Local Area Network

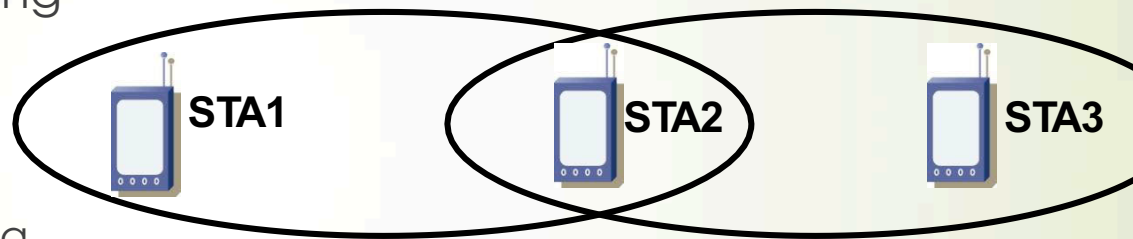


Technology Advances in Wireless Communication/Networks

- Physical: Lower power consumption, faster processor, larger battery (**but still not enough**)
- Link: Better/more antenna, more **aggressive** modulation and coding schemes
- Network: Mobility support, resource allocation (**unicast/multicast**)
- Application: Adaptive QoS
- Overall: **Software-Defined Radio Networks**
 - Cognitive Radio
 - Cloud Radio Access Networks (CRANs)

General Challenges in Wireless Networks

- Hidden terminal problem
- Eavesdropping
- Reliability
 - Fading
 - Shadowing
 - Multipath
 - Interference (co-channel, inter-channel)
- Power control/consumption
 - Q: Why do we need power control?
- Limited wireless spectrum: licensed versus unlicensed (ISM band, industrial, scientific, and medical)



Q: ISM band is widely used nowadays, what is the side effect?

5

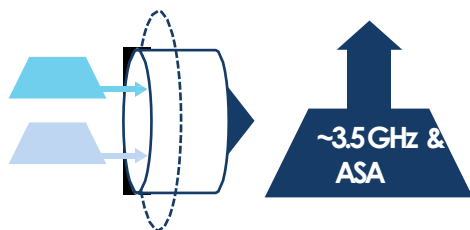
How Can We Increase Capacity

Shannon's Law

$$C \approx W \cdot n \cdot \log_2(1 + SNR)$$

Capacity Spectrum Antennas Signal Quality

More Spectrum



Making the best use of all spectrum types with more licensed spectrum as the top priority, e.g., ASA, ~3.5 GHz, unlicensed spectrum

More Antennas



Advanced multiple antenna techniques to create spatially separated data paths, e.g., 4 way receive diversity, 4x4 MIMO

Interference Mitigation



Advanced receivers and antenna techniques, e.g., LTE FeICIC/IC, HSPA+ advanced device receiver

WiFi is Probably and Most Widely

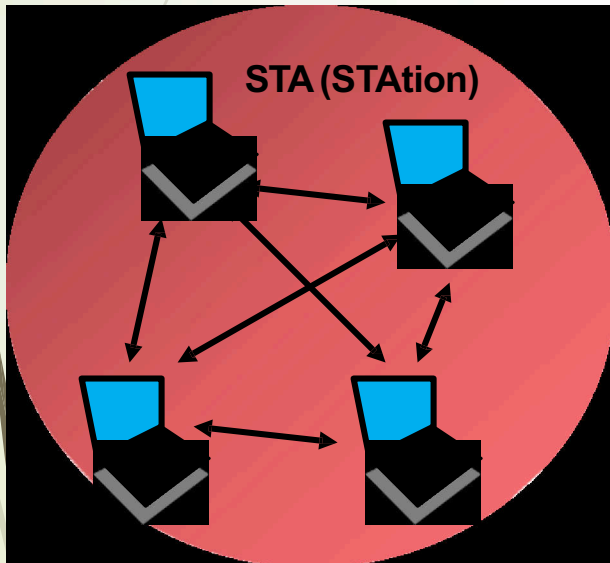
Note: 802.11 standards are more than radios!

- Low cost WLAN solution over ISM band
- 802.11 standard family:
 - IEEE 802.11a: 54 Mbit/s, 5 GHz standard (1999, shipping products in 2001)
 - IEEE 802.11b: Enhancements to 802.11 to support 5.5 Mbit/s and 11 Mbit/s (1999)
 - IEEE 802.11e: Enhancements: QoS, including packet bursting (2005)
 - IEEE 802.11g: 54 Mbit/s, 2.4 GHz standard (backwards compatible with b) (2003)
 - IEEE 802.11n: Higher-throughput improvements using MIMO (multiple-input, multiple-output antennas) (September 2009)
 - IEEE 802.11ac: Very High Throughput <6 GHz; potential improvements over 802.11n: better modulation scheme (expected ~10% throughput increase), wider channels (estimate in future time 80 to 160 MHz), multi user MIMO (December 2013)

Two WiFi Modes

Ad-hoc mode:

No access points; STAs communicate directly with each other.



IBSS:
Independent Basic Service Set

Infrastructure mode:

Usage of access points interconnected with wired LAN.

